EXPERIMENT 7

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EXPERIMENT 7
Design and Performance Analysis of Materialized Views (SanDisk, JTG, PayPal)

AIM
To design and implement a materialized view and to compare and analyze execution time and performance differences between simple views, complex views, and materialized views, thereby understanding their impact on query optimization and system performance.

OBJECTIVE

1. **To create simple views, complex views, and materialized views.**
2. **To evaluate their performance by comparing query execution time.**
3. **To highlight the advantages of materialized views in enterprise-level applications.**

SOFTWARE REQUIREMENTS

Database Management System:
Oracle Database Express Edition (Oracle XE)
PostgreSQL Database

Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

PROBLEM STATEMENT

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In large-scale enterprise systems, frequent execution of complex queries can significantly affect performance. Organizations require optimized mechanisms to improve query response time without compromising data consistency.

Materialized views help improve performance by storing precomputed query results, reducing the need for repeated complex calculations.

PRACTICAL / EXPERIMENT STEPS

Step 1: Create base tables (Employee, Department, Payroll).
Step 2: Insert sample data into tables.
Step 3: Create a simple view based on a single table.
Step 4: Create a complex view involving joins and aggregation.
Step 5: Create a materialized view storing precomputed results.
Step 6: Execute queries on all views.
Step 7: Compare execution time and analyze performance.

STEP-WISE OUTPUT

Step 1 – Create Base Tables

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[SQL Worksheet]*

```sql
-- Employee Table
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptID INT,
    Salary NUMBER(10,2)
);

-- Department Table
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(50)
);

-- Payroll Table
CREATE TABLE Payroll (
    EmpID INT,
    PayDate DATE,
    PayAmount NUMBER(10,2),
    PRIMARY KEY (EmpID, PayDate),
    FOREIGN KEY (EmpID) REFERENCES Employee(EmpID)
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Table DEPARTMENT created.

Elapsed: 00:00:00.011

SQL> CREATE TABLE Payroll (
         EmpID INT,
         PayDate DATE,
         PayAmount NUMBER(10,2),...
Show more...
```

Step 2 – Insert Sample Data

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[SQL Worksheet]*

```sql
-- Departments
INSERT INTO Department VALUES (1, 'IT');
INSERT INTO Department VALUES (2, 'Finance');
INSERT INTO Department VALUES (3, 'HR');

-- Employees
INSERT INTO Employee VALUES (101, 'Alice', 1, 70000);
INSERT INTO Employee VALUES (102, 'Bob', 2, 80000);
INSERT INTO Employee VALUES (103, 'Charlie', 1, 65000);
INSERT INTO Employee VALUES (104, 'David', 3, 60000);

-- Payroll Records
INSERT INTO Payroll VALUES (101, TO_DATE('2026-02-01','YYYY-MM-DD'), 7000);
INSERT INTO Payroll VALUES (102, TO_DATE('2026-02-01','YYYY-MM-DD'), 8000);
INSERT INTO Payroll VALUES (103, TO_DATE('2026-02-01','YYYY-MM-DD'), 6500);
INSERT INTO Payroll VALUES (104, TO_DATE('2026-02-01','YYYY-MM-DD'), 6000);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
1 row inserted.

Elapsed: 00:00:00.002

SQL> INSERT INTO Payroll VALUES (103, TO_DATE('2026-02-01','YYYY-MM-DD'), 6500)


1 row inserted.

Elapsed: 00:00:00.002
```

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Step 3 – Create Simple View

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[SQL Worksheet]*

```
CREATE VIEW SimpleEmployeeView AS
SELECT EmpID, EmpName, Salary
FROM Employee;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE VIEW SimpleEmployeeView AS
     SELECT EmpID, EmpName, Salary
     FROM Employee


View SIMPLEEMPLOYEEVIEW created.

Elapsed: 00:00:00.014
```

Step 4 – Create Complex View

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[SQL Worksheet]*

```sql
CREATE VIEW ComplexEmployeePayroll AS
SELECT e.EmpID, e.EmpName, d.DeptName, SUM(p.PayAmount) AS TotalPaid
FROM Employee e
JOIN Department d ON e.DeptID = d.DeptID
JOIN Payroll p ON e.EmpID = p.EmpID
GROUP BY e.EmpID, e.EmpName, d.DeptName;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE VIEW ComplexEmployeePayroll AS
     SELECT e.EmpID, e.EmpName, d.DeptName, SUM(p.PayAmount) AS TotalPaid
     FROM Employee e
     JOIN Department d ON e.DeptID = d.DeptID...
Show more...


View COMPLEXEMPLOYEEPAYROLL created.

Elapsed: 00:00:00.016
```

Step 5 – Create Materialized View

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[SQL Worksheet]*

```sql
CREATE MATERIALIZED VIEW MaterializedEmployeePayroll
BUILD IMMEDIATE
REFRESH COMPLETE ON DEMAND
AS
SELECT e.EMPID, e.EMPNAME, d.DEPTNAME, SUM(p.PAYAMOUNT) AS TOTALPAI
FROM Employee e
JOIN Department d ON e.DEPTID = d.DEPTID
JOIN Payroll p ON e.EMPID = p.EMPID
GROUP BY e.EMPID, e.EMPNAME, d.DEPTNAME;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE MATERIALIZED VIEW MaterializedEmployeePayroll
     BUILD IMMEDIATE
     REFRESH COMPLETE ON DEMAND
     AS...
Show more...
```

Step 6 – Execute Queries on Views

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[SQL Worksheet]*

```sql
-- Simple View Query
SELECT * FROM SimpleEmployeeView;

-- Complex View Query
SELECT * FROM ComplexEmployeePayroll;

-- Materialized View Query
SELECT * FROM MaterializedEmployeePayroll;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	EMPID	EMPNAME	SALARY
1	101	Alice	70000
2	102	Bob	80000
3	103	Charlie	65000
4	104	David	60000

Step 7 – Compare Performance

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[SQL Worksheet]*

```sql
SET TIMING ON;

SELECT * FROM SimpleEmployeeView;
SELECT * FROM ComplexEmployeePayroll;
SELECT * FROM MaterializedEmployeePayroll;

SET TIMING OFF;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.011 seconds

	EMPID	EMPNAME	SALARY
1	101	Alice	70000
2	102	Bob	80000
3	103	Charlie	65000
4	104	David	60000

RESULT ANALYSIS

1. **Simple View**
   - **Created from a single table.**
   - **Executes quickly for small datasets.**
   - **No performance optimization for complex operations.**
2. **Complex View**
   - **Includes joins and aggregation.**
   - **Slower than simple view because data is computed every time the query runs.**

3. **Materialized View**
    - **Stores precomputed results physically.**
    - **Significantly faster for repeated queries.**
    - **Requires refresh mechanism to maintain updated data.**

REAL-WORLD APPLICATION

Companies such as SanDisk, JTG, and PayPal handle large-scale transactional data.

In such organizations:

- **Complex reports run frequently.**
- **Data warehouses require fast query responses.**
- **Business intelligence dashboards need optimized performance.**

Materialized views are used to:

- **Improve reporting speed.**
- **Reduce server workload.**
- **Enhance query optimization.**
- **Improve overall system performance.**

LEARNING OUTCOME

After completing this experiment, the learner is able to:

1. **Understand the concept and working of materialized views in database systems.**
2. **Differentiate between simple views, complex views, and materialized views.**
3. **Create and refresh materialized views in Oracle and PostgreSQL.**
4. **Measure and compare query execution time.**
5. **Analyze performance benefits in data-intensive enterprise applications.**

CONCLUSION

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The experiment demonstrates that materialized views significantly improve query performance for repeated complex operations. While simple and complex views calculate results dynamically, materialized views store results physically, reducing computation time.

Therefore, materialized views are highly suitable for enterprise-level systems where performance and efficiency are critical.